MBFT 814 : Microbial Biotechnology 3(2-0-2)

M.Sc. Microbiology, Semester III · Mid-term notes

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Topic	What it covers
1. Introduction to microbial biotechnology	Definitions, why microbes, scope, primary vs secondary metabolites
2. History of microbial biotechnology	Eras, timeline table, key scientists
3. Types of bioreactors	Parts, CSTR, bubble column, airlift, fluidized bed, packed bed, membrane, photobioreactor
4. Types of fermentation methods	Batch, fed-batch, continuous, submerged, surface, solid state, semi-solid
5. Enzyme production	Microbial sources, steps, downstream, applications
6. Immobilization of enzymes	Adsorption, covalent, entrapment, encapsulation, cross-linking
7. Screening and strain improvement	Primary/secondary screening, mutation, recombination, rDNA
8. Antibiotic production	Penicillin (detail), streptomycin, tetracycline, griseofulvin
9. Vitamin production	Riboflavin (B2), cyanocobalamin (B12)
Quick revision + likely questions	Last page

1. INTRODUCTION TO MICROBIAL BIOTECHNOLOGY

Definitions

- **Biotechnology**: the term was coined by **Karl Ereky** (Hungarian engineer) in **1919**, meaning production of products from raw materials with the help of living organisms.
- **OECD / CBD definition**: "Any technological application that uses biological systems, living organisms or derivatives thereof, to make or modify products or processes for specific use."
- Also defined as "the integrated use of **microbiology, biochemistry and engineering sciences** in production or as a service operation".
- **Microbial biotechnology**: use of **microorganisms** (bacteria, fungi, yeasts, algae) or their products (enzymes, metabolites) on an industrial scale to make useful products or provide services such as waste treatment.

Why microorganisms are preferred

- Very **fast growth** (generation time minutes to hours) and high surface-to-volume ratio, so high metabolic rate.
- Grow on **cheap substrates** and agro-industrial wastes (molasses, corn steep liquor, whey, bran).
- **Huge metabolic diversity**: can make alcohols, acids, enzymes, antibiotics, vitamins, polymers.
- Easy to **genetically manipulate** (mutation, rDNA) and easy to **scale up** in fermenters.
- Production is independent of climate and season; product quality is uniform.

Scope / applications

Area	Examples
Food and beverages	Beer, wine, bread, cheese, yoghurt, vinegar, SCP, amino acids (glutamic acid, lysine)
Pharmaceuticals	Antibiotics, vitamins, vaccines, insulin, interferons, steroids (biotransformation), enzymes
Industrial chemicals	Ethanol, acetone–butanol, glycerol, citric, lactic and gluconic acid, biopolymers
Enzymes	Amylase, protease, lipase, cellulase, pectinase, glucose isomerase
Agriculture	Biofertilizers (Rhizobium, Azotobacter, PSB), biopesticides (Bacillus thuringiensis), silage

Area	Examples
Environment	Sewage treatment, bioremediation of oil and pesticides, biogas, bioleaching of metals
Energy	Bioethanol, biogas (methane), biohydrogen, microbial fuel cells

Primary vs secondary metabolites (very commonly asked)

Primary metabolites	Secondary metabolites
Produced during active growth (**trophophase**, log phase)	Produced after growth slows (**idiophase**, late log / stationary)
Essential for growth	Not essential for growth
Common to most organisms	Specific to few organisms
e.g. ethanol, amino acids, citric acid, vitamins, enzymes	e.g. antibiotics, toxins, pigments, alkaloids

Exam point: Define biotechnology with Karl Ereky 1919, then write 4 reasons microbes are used and a scope table. That covers a 5-mark answer.

2. HISTORY OF MICROBIAL BIOTECHNOLOGY

Eras of biotechnology

- **Ancient biotechnology** (before 2000 BC): food, shelter, domestication; fermented foods by chance.
- **Classical biotechnology** (2000 BC to 1800s AD): fermentation for food and medicine; beer, wine, cheese, vinegar.
- **1900–1953**: genetics, industrial fermentations (acetone–butanol, citric acid), antibiotics.
- **1953–1976**: DNA structure and molecular biology.
- **Modern biotechnology** (1977 onwards): genetic engineering, recombinant products, genomics.

Timeline of important events

Year	Event
6000 BC	Sumerians and Babylonians used yeast to make **beer**
4000 BC	Egyptians made **leavened bread** with yeast; cheese and vinegar also ancient
1673–1676	**Antonie van Leeuwenhoek** first saw microorganisms ("animalcules") with his microscope
1796	Edward Jenner: smallpox vaccination with cowpox (Pylarini had described inoculation in 1701)
1809	Nicolas Appert: canning and heat sterilization of food
1857–1876	**Louis Pasteur** proved fermentation is caused by living microbes (lactic 1857, alcoholic); pasteurization; "father of industrial microbiology"
1877	Kühne coined the word "enzyme"
1897	**Eduard Buchner**: cell-free yeast extract ferments sugar, so enzymes cause fermentation (Nobel 1907)
1914–1918	**Chaim Weizmann**: acetone–butanol fermentation by *Clostridium acetobutylicum* from starch; acetone used for explosives (cordite) in World War I
1919	**Karl Ereky** coined the term "biotechnology"
1923	Pfizer: industrial **citric acid** by *Aspergillus niger* (surface culture)
1928	**Alexander Fleming** discovered penicillin from *Penicillium notatum*
1938–1945	**Florey and Chain** purified penicillin; large-scale **submerged (deep-tank) fermentation** developed at Peoria, USA using corn steep liquor and *P. chrysogenum*
1944	**Selman Waksman** (with Schatz) discovered **streptomycin** from *Streptomyces griseus*, first drug against TB
1950s	Microbial biotransformation of steroids (cortisone) on industrial scale
1953	**Watson and Crick**: double-helix structure of DNA

Year	Event
1969–1973	First industrial use of an immobilized enzyme (aminoacylase, Tanabe, Japan); **Cohen and Boyer** made the first recombinant DNA (1973)
1977	**Genentech** produced **somatostatin** in bacteria: first human protein from a cloned gene
1978–1982	Human **insulin** cloned in *E. coli* (1978); approved as Humulin in 1982, first rDNA drug
1983	Kary Mullis invented **PCR**
1986	First recombinant vaccine approved: **hepatitis B** (made in yeast)
1998	Remicade, a monoclonal antibody for Crohn's disease, approved

Exam point: Key names: Leeuwenhoek (microbes), Pasteur (fermentation by microbes), Buchner (enzymes, cell-free), Weizmann (acetone–butanol, WWI), Ereky (term 1919), Fleming (penicillin 1928), Waksman (streptomycin 1944), Cohen–Boyer (rDNA 1973).

3. TYPES OF BIOREACTORS

A **bioreactor (fermenter)** is a vessel that provides a controlled, sterile environment (temperature, pH, oxygen, mixing, nutrients) for microorganisms, cells or enzymes to carry out a biochemical process and form a product.

Basic parts of a stirred fermenter

- **Vessel**: stainless steel (large) or glass (lab); height:diameter ratio usually 2–3 : 1.
- **Agitator / impeller** (Rushton disc turbine) on a motor-driven shaft for mixing and breaking air bubbles.
- **Sparger**: introduces sterile air at the bottom (porous, orifice or nozzle type).
- **Baffles**: 4 metal strips on the wall to stop vortex formation and improve mixing.
- **Cooling jacket or coils** for temperature control.
- **Probes and controls**: temperature, pH, dissolved oxygen (DO), foam sensor with antifoam addition, pressure gauge.
- **Ports**: inoculation, sampling, feeding, harvest valve; air filter and exhaust with condenser.

Types and their principle

Bioreactor	Principle / design	Advantages	Uses
Continuous stirred tank (CSTR)	Cylindrical vessel with impellers, baffles and sparger; mechanical mixing	Good mixing and O_2 transfer, flexible, most common	Antibiotics, enzymes, most aerobic fermentations
Bubble column	Tall column, air sparged at bottom; rising bubbles mix the medium; no impeller	Simple, cheap, low shear, no moving parts	Baker's yeast, beer, vinegar, SCP
Airlift	Draft tube or baffle divides vessel into **riser** (gassed) and **downcomer** (ungassed); density difference drives circulation. Internal-loop or external-loop	Low shear, low energy, good for shear-sensitive cells, easy scale-up	SCP (ICI Pruteen), animal/plant cells, wastewater
Fluidized bed	Like bubble column but **expanded top** reduces fluid velocity; immobilized cells/particles kept suspended by upward flow	Good mass transfer, no clogging, suits immobilized cells	Wastewater treatment, beer, vinegar, immobilized enzymes
Packed bed	Column packed with solid support carrying immobilized cells/enzymes; medium flows through the bed	High cell density, simple, continuous	Immobilized enzymes (glucose isomerase), vinegar (trickle), effluent treatment
Membrane / hollow fibre	Cells held on one side of semi-permeable membrane fibres; nutrients in and product out by diffusion	Very high cell density, product separated continuously, low shear	Animal cells, monoclonal antibodies, enzymes

Bioreactor	Principle / design	Advantages	Uses
Photobioreactor	Transparent vessel (tubular, flat panel) or open raceway pond with light source for photosynthetic organisms	Controlled light and CO_2, less contamination (closed type)	Spirulina, Chlorella, algal biofuel, pigments

Classification by mode of operation

- **Batch**, **fed-batch**, **continuous** (chemostat, turbidostat), and **continuous with cell recycle** (cells returned to keep density high). Details in Topic 4.

Exam point: Draw a labelled stirred-tank fermenter (motor, shaft, impeller, baffles, sparger, jacket, probes, ports). For airlift, draw the draft tube with riser and downcomer.

4. TYPES OF FERMENTATION METHODS

A. Based on operation

Type	What happens	Advantages	Disadvantages / example
Batch (closed system)	All nutrients added at start; nothing added or removed except air, acid/base and antifoam; harvested at end. Cells pass through lag, log, stationary, death phases	Simple, low contamination risk, suits secondary metabolites	Down-time for cleaning and sterilizing, low productivity; e.g. beer, many antibiotics
Fed-batch	Nutrients fed continuously or step-wise during run, product removed at end; volume increases	Avoids **substrate inhibition** and **catabolite repression**, extends production phase, high yield	Needs feed control; e.g. **penicillin**, baker's yeast
Continuous (open system)	Fresh medium added and broth removed at the same rate; volume constant; cells kept in log phase at **steady state**	High productivity, uniform product, less down-time	High contamination risk, strain mutation; e.g. SCP, ethanol, wastewater
Chemostat	Growth controlled by a **limiting nutrient**; at steady state specific growth rate μ = dilution rate D (D = F/V)	Stable, easy to study kinetics	Washout if D > μmax
Turbidostat	Cell density (turbidity) kept constant by adjusting feed through a photocell	Works near μmax	Complex control

B. Based on physical state of medium

Type	Description	Examples
Submerged (SmF)	Microbes grow inside liquid medium with aeration and agitation; most industrial processes	Antibiotics, enzymes, citric acid (modern), vitamins
Surface fermentation	Organism grows as a mat on the surface of a still liquid medium in shallow trays	Early citric acid by *A. niger*, vinegar (Orleans)
Solid state (SSF)	Growth on moist **solid substrate** with little or no free water (moisture about 40–80%); mainly fungi	Koji, enzymes (amylase, cellulase) on wheat bran, tempeh, mushrooms, compost
Semi-solid	Between SmF and SSF; substrate is a thick slurry with some free water	Lipase and enzyme production on bran slurries (syllabus practical)

SmF vs SSF	Submerged	Solid state
Water	Plenty of free water	Little or no free water
Organisms	Bacteria, yeast, fungi	Mainly filamentous fungi
Control	Easy (pH, temp, O_2)	Difficult (heat build-up, gradients)
Cost / waste	More energy and water, more effluent	Cheap agro-waste substrates, less effluent, concentrated product

C. Other classifications

- **Aerobic** (antibiotics, citric acid) vs **anaerobic** (ethanol, lactic acid, acetone–butanol, biogas).
- **Single-stage** vs **multi-stage**; pure culture vs mixed culture.

Exam point: Draw the batch growth curve (lag, log, stationary, death). Write "at steady state μ = D" for chemostat and explain washout.

5. ENZYME PRODUCTION

Why microbial enzymes

- Grow fast in large amounts; easy to extract (many are **extracellular**).
- Cheap media, no seasonal variation, easy to improve strains.
- Wide range of enzymes, many stable at high temperature and pH.

Steps in microbial enzyme production

1. **Selection of organism**: high yield, GRAS status (safe), stable strain, non-toxic, short fermentation.
2. **Medium formulation**: carbon (starch, molasses), nitrogen (soybean meal, corn steep liquor, ammonium salts), minerals and **inducer** (e.g. starch for amylase, oil for lipase, cellulose for cellulase). Avoid catabolite repression by glucose.
3. **Inoculum preparation**: slant, then shake flask, then seed fermenter (about 5–10% v/v).
4. **Fermentation**: submerged (most bacterial enzymes) or solid state (many fungal enzymes); control pH, temperature, aeration; enzyme usually peaks in late log or early stationary phase.
5. **Downstream processing**: (a) remove cells by centrifugation or filtration; (b) for intracellular enzymes, **cell disruption** (homogenizer, bead mill, sonication, enzymatic lysis, osmotic shock); (c) concentration by ultrafiltration or vacuum evaporation; (d) **precipitation** with ammonium sulphate or acetone/ethanol; (e) purification by dialysis and chromatography (ion exchange, gel filtration, affinity) if high purity is needed.
6. **Formulation**: stabilizers (glycerol, sorbitol, salts), preservatives; sold as liquid, powder or granules; enzyme activity standardized.

Important microbial enzymes and applications

Enzyme	Source organism	Application
α-Amylase	*Bacillus amyloliquefaciens*, *B. licheniformis*, *Aspergillus oryzae*	Starch liquefaction, baking, textile desizing, detergent
Glucoamylase	*Aspergillus niger*	Glucose syrup from starch, brewing
Protease (subtilisin)	*Bacillus licheniformis*, *B. subtilis*	Detergents, leather, meat tenderizing
Lipase	*Candida rugosa*, *Rhizopus*, *Aspergillus*	Detergents, dairy flavour, biodiesel
Cellulase	*Trichoderma reesei*	Bioethanol, textile (stone-washing), juice
Pectinase	*Aspergillus niger*	Fruit juice clarification, wine
Glucose isomerase	*Streptomyces* spp.	High fructose corn syrup (HFCS)
Microbial rennet / chymosin	*Mucor miehei*; recombinant chymosin	Cheese making
Penicillin acylase	*E. coli*	6-APA for semi-synthetic penicillins
L-Asparaginase	*E. coli*, *Erwinia*	**Therapeutic**: acute lymphoblastic leukaemia
Streptokinase	*Streptococcus* (β-haemolytic)	**Therapeutic**: dissolves blood clots
Glucose oxidase	*Aspergillus niger*, *Penicillium*	**Analytical**: blood glucose biosensor strips

- **Therapeutic uses**: asparaginase (leukaemia), streptokinase and urokinase (clots), lysozyme, digestive enzymes.
- **Analytical uses**: glucose oxidase and peroxidase (glucose estimation, biosensors), HRP and alkaline phosphatase in ELISA, restriction enzymes and Taq polymerase in molecular biology.
- **Industrial uses**: detergent, starch, food, dairy, brewing, textile, leather, paper.

6. IMMOBILIZATION OF ENZYMES

Immobilization is the physical confinement or localization of an enzyme (or whole cell) on or inside a support (matrix) so that it keeps its catalytic activity and can be **used again and again** or continuously.

Advantages

- Enzyme can be **reused** and used in **continuous** processes (packed bed, fluidized bed).
- Product is free of enzyme, so easy downstream processing.
- Often **more stable** to heat, pH and storage.
- Reaction can be controlled or stopped by removing the enzyme; lower cost per unit product.

Limitations

- Some **loss of activity** during immobilization.
- Diffusion (mass transfer) limitation, especially for large substrates.
- Cost of support and process; enzyme may leak from matrix.

Methods of immobilization

Method	Principle	Supports / agents	Merit / demerit
1. Adsorption	Enzyme binds to surface by weak forces (van der Waals, ionic, hydrogen bonds)	Activated charcoal, alumina, silica, ion-exchange resins (DEAE-cellulose), glass	Simple, cheap, little activity loss / enzyme can leak (desorption)
2. Covalent bonding	Covalent bond between enzyme groups (-NH_2, -COOH) and activated support	CNBr-activated agarose/Sepharose, cellulose, porous glass with carbodiimide or glutaraldehyde	Strong, no leakage / may damage active site, costly
3. Entrapment	Enzyme or cells trapped inside the lattice of a polymer gel; substrate diffuses in	**Calcium alginate** beads, polyacrylamide, agar, gelatin, carrageenan	Mild, suits whole cells / leakage of small enzymes, diffusion limit
4. Encapsulation (microencapsulation)	Enzyme enclosed inside semi-permeable membrane capsules or liposomes	Nylon, cellulose nitrate membranes, liposomes, hollow fibres	Large enzyme load / only small substrates pass
5. Cross-linking (carrier-free)	Enzyme molecules joined to each other by bifunctional reagents	**Glutaraldehyde**; CLEA (cross-linked enzyme aggregates)	No support needed / can lose activity, gel is weak

Applications of immobilized enzymes / cells

- **Glucose isomerase** (immobilized) for high fructose corn syrup: the largest industrial use.
- **Aminoacylase** for resolution of L-amino acids: first industrial immobilized enzyme (Tanabe Seiyaku, Japan, 1969).
- **Penicillin acylase** to make 6-APA for semi-synthetic penicillins.
- **Lactase** (β-galactosidase) for lactose-free milk; **biosensors** (glucose oxidase electrode); immobilized yeast for ethanol and beer.

Exam point: Classify the five methods with one diagram each (beads for entrapment, membrane circle for encapsulation, enzyme on surface for adsorption, link lines for covalent and cross-linking).

7. SCREENING AND STRAIN IMPROVEMENT

Screening

Screening is the set of highly selective procedures used to detect and isolate the few useful organisms (producing a desired product) from a very large number of isolates. Sources: soil, water, compost, decaying matter, culture collections (ATCC, MTCC, NCIM).

Primary screening (qualitative, find producers)

- **Crowded plate technique**: soil dilution plated so many colonies grow; colonies with a clear zone around them produce antibiotics.
- **Auxanography**: detects organisms producing growth factors (amino acids, vitamins) using a test organism that needs them.
- **Indicator dye / zone methods**: pH indicator for acid producers (e.g. bromocresol green for organic acids), starch–iodine for amylase, skim milk for protease, tributyrin for lipase, CMC–Congo red for cellulase.
- **Enrichment culture**: medium favouring the desired type (e.g. only the target substrate as carbon source).
- **Giant colony technique** and **agar plug / cross-streak** methods for antibiotic spectrum.

Secondary screening (quantitative, pick the best)

- Done in **shake flasks** or small fermenters; measures actual **yield**, productivity and fermentation time.
- Identifies the organism and checks if product is new or known; checks stability, toxicity, cost of medium.
- Studies optimum pH, temperature, aeration, and whether the product is intracellular or extracellular.

Strain improvement

Wild strains usually give low yield. **Strain improvement** is the genetic alteration of an organism to increase product yield, reduce by-products, use cheaper media, or improve stability.

Approach	Method	Notes / example
1. Mutation and selection (classical)	Physical mutagens: **UV**, X-rays, gamma rays. Chemical: **NTG** (N-methyl-N′-nitro-N-nitrosoguanidine), **EMS** (ethyl methane sulphonate), nitrous acid, base analogues	Random; thousands of survivors screened. Main method for penicillin improvement
Selection of mutants	Auxotrophic mutants, **regulatory mutants** resistant to feedback inhibition (analogue-resistant), revertants	e.g. lysine overproducers of *Corynebacterium glutamicum*
2. Recombination	Protoplast fusion, parasexual cycle in fungi, conjugation, transduction, transformation	Combines good traits of two strains
3. Recombinant DNA technology	Cloning extra copies of genes, strong promoters, expressing genes in new hosts	Insulin in *E. coli*, chymosin in yeast
4. Site-directed mutagenesis / protein engineering	Specific change in a known gene	More stable detergent enzymes
5. Metabolic engineering	Remove competing pathways, remove feedback regulation	Riboflavin in *B. subtilis*

Classic example: penicillin strain improvement

P. notatum (Fleming) gave very low yield → *P. chrysogenum* **NRRL 1951** (from a mouldy cantaloupe, Peoria) → X-ray mutant **X-1612** → UV mutant **Wis Q-176** (Wisconsin) → later mutants. Yield rose from a few units/mL to over **50,000 units/mL**.

Preservation of improved strains

- Subculturing on slants and storage at 4°C; overlay with mineral oil.
- **Lyophilization** (freeze-drying); storage in **liquid nitrogen** (−196°C) or glycerol stocks at −80°C.

8. ANTIBIOTIC PRODUCTION

Antibiotics are secondary metabolites produced by microorganisms that, in low concentration, inhibit or kill other microorganisms. They are made in the **idiophase** by aerobic submerged, usually **fed-batch**, fermentation.

A. Penicillin (detailed)

Point	Details
Organism	*Penicillium chrysogenum* (high-yield mutants); discovered by Fleming in 1928 from *P. notatum*
Medium	Carbon: **lactose** (slowly used, avoids glucose repression) or glucose fed slowly; nitrogen: **corn steep liquor**; minerals; $CaCO_3$ as buffer; **precursor phenylacetic acid** (for penicillin G) or phenoxyacetic acid (for penicillin V), fed in small amounts because it is toxic; antifoam
Inoculum	Spores → shake flasks → seed tank; about 5–10% inoculum
Conditions	Aerobic, stirred tank, **fed-batch**; temperature about **25–27°C**; pH about **6.5** (kept 6.4–6.8); high aeration; about 5–7 days
Phases	**Trophophase** (first 1–2 days, mycelial growth, pH falls) then **idiophase** (penicillin made, feeding of sugar and precursor, pH controlled)
Recovery	Filter mycelium (rotary vacuum filter) → cool and acidify filtrate to **pH 2.0–2.5** → extract into solvent (**butyl acetate / amyl acetate**) → back-extract into aqueous buffer at pH 7.5 → treat with charcoal → crystallize as **potassium or sodium salt** → dry, test
Semi-synthetic	Penicillin G + **penicillin acylase** → **6-APA** → add side chains → ampicillin, amoxicillin, methicillin

B. Streptomycin

- Organism: *Streptomyces griseus*; discovered by **Waksman** (with Schatz) in 1944; aminoglycoside, active against Gram-negative bacteria and *Mycobacterium tuberculosis*.
- Medium: glucose, **soybean meal** (N source), NaCl; aerobic submerged, about 28°C, pH 7.6–8.0, about 7–10 days.
- Three phases: (1) rapid growth with release of NH_3 (pH rises); (2) streptomycin production, glucose used, pH stable; (3) sugar exhausted, cells lyse.
- Recovery: remove mycelium, adsorb on **cation exchange resin** (or activated charcoal), elute with acid, precipitate and crystallize.

C. Tetracyclines

- Chlortetracycline (aureomycin) from *Streptomyces aureofaciens*; oxytetracycline (terramycin) from *S. rimosus*.
- Broad-spectrum, inhibit protein synthesis (30S). Aerobic, about 28°C.
- Important point: for **tetracycline** (not chlortetracycline), **chloride is kept low** or a chlorination inhibitor (bromide) is added, so that chlorination does not occur. Phosphate kept low as it represses production.

D. Griseofulvin

- Antifungal antibiotic from *Penicillium griseofulvum* (also *P. patulum*); used orally for ringworm (dermatophytes).
- Aerobic submerged fermentation; **chloride** must be added because griseofulvin contains a chlorine atom; recovered by solvent extraction.

Exam point: For penicillin write: organism, medium with precursor, fed-batch conditions, two phases and recovery steps as a flow chart. Most asked production question.

9. VITAMIN PRODUCTION

A. Riboflavin (vitamin B_2)

Point	Details
Organisms	***Ashbya gossypii*** and *Eremothecium ashbyii* (fungi, natural overproducers); *Candida famata* (yeast); recombinant ***Bacillus subtilis***
Medium	Glucose or plant oils, corn steep liquor, peptone, collagen/gelatin; **glycine** as precursor; low iron needed for *Candida*
Conditions	Aerobic submerged, about 26–28°C, pH around 6–7, about 5–7 days; riboflavin made in stationary phase and partly crystallizes
Recovery	Heat broth to release riboflavin from cells, separate cells, crystallize and dry
Uses	Animal feed, food colour (yellow), vitamin supplements

B. Cyanocobalamin (vitamin B_{12})

Point	Details
Organisms	***Propionibacterium freudenreichii*** (and *P. shermanii*), ***Pseudomonas denitrificans***. Made only by bacteria, not by plants or animals
Medium	Glucose (or beet molasses), corn steep liquor; **cobalt** salt (B_{12} contains cobalt) and precursor **5,6-dimethylbenzimidazole (DMB)**
Propionibacterium process	**Two stages**: first 2–4 days **anaerobic** (growth and cobinamide formation), then 3–4 days **aerobic** (DMB is made and joined, forming B_{12})
Pseudomonas process	Aerobic throughout; betaine from beet molasses increases yield
Recovery	B_{12} is intracellular: cells harvested, heated to release it, **cyanide** added to convert all forms to stable **cyanocobalamin**, then purified and crystallized
Uses	Treatment of pernicious anaemia, food and feed supplement

Other vitamin (for extra marks): Vitamin C (ascorbic acid) by the Reichstein process uses *Gluconobacter oxydans* to oxidize sorbitol to sorbose; β-carotene (provitamin A) from *Blakeslea trispora* and algae *Dunaliella*.

10. QUICK REVISION AND LIKELY QUESTIONS

Remember	Answer
Term "biotechnology"	Karl Ereky, 1919
Fermentation is caused by microbes	Louis Pasteur
Cell-free fermentation (enzymes)	Eduard Buchner, 1897
Acetone–butanol	Chaim Weizmann, *Clostridium acetobutylicum*, WWI
Penicillin / streptomycin	Fleming 1928 / Waksman 1944
First rDNA	Cohen and Boyer 1973; somatostatin 1977; insulin 1982 (approved)
Airlift zones	Riser (gassed) and downcomer
Chemostat steady state	$\mu = D$ ($D = F/V$); washout if $D > \mu max$
Fed-batch used for	Penicillin, baker's yeast; avoids substrate inhibition and catabolite repression
SSF	Little free water, fungi, wheat bran, koji
Commonest entrapment matrix	Calcium alginate
Cross-linking agent	Glutaraldehyde
Largest immobilized enzyme use	Glucose isomerase → HFCS
Chemical mutagens	NTG, EMS, nitrous acid
Penicillin precursor	Phenylacetic acid (Pen G)
Penicillin extraction pH / solvent	pH 2–2.5, butyl or amyl acetate
B_{12} organism / metal	*Propionibacterium*, *Pseudomonas denitrificans* / cobalt + DMB
Riboflavin organism	*Ashbya gossypii*, *Eremothecium ashbyii*, *B. subtilis*

Likely questions

1. Define biotechnology. Give the history of microbial biotechnology with important milestones.
2. Describe the design of a stirred tank fermenter with a labelled diagram.
3. Write notes on airlift, bubble column, fluidized bed, packed bed and photobioreactors.
4. Differentiate batch, fed-batch and continuous fermentation. Add a note on chemostat.
5. Compare submerged and solid state fermentation.
6. Describe the methods of enzyme production and give industrial, therapeutic and analytical applications of enzymes.
7. What is enzyme immobilization? Describe its methods with advantages and disadvantages.
8. Explain primary and secondary screening and the methods of strain improvement.
9. Describe the industrial production of penicillin.
10. Write short notes: streptomycin, tetracycline, griseofulvin, riboflavin, vitamin B_{12}.

References: Casida, Industrial Microbiology; Patel, Industrial Microbiology; Stanbury, Whitaker and Hall, Principles of Fermentation Technology; class notes "History of Biotechnology" and "Bioreactor Design" (Unit 5, Dr. Annam Renita A.).